

Vinitha Gadiraju

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Faculty Position

Wellesley College

July 2023 - Present

Assistant Professor in Computer Science, Tenure-Track

Education

University of Colorado, Boulder (CU Boulder)

August 2018 - May 2023

Ph.D. Computer Science

M.S. Computer Science

University of Oregon

September 2015 - June 2018

B.S. Computer Science, Minor in Psychology

Awards

- 2025 **Trustworthy AI Research Fellow**, Computing Research Association and Microsoft (\$17,500)
 - 2024 **Research Scholar Program Award**, Google Research (\$60,000)
 - 2022 **Outstanding Service Award**, CU Boulder, Computer Science Department (\$600)
 - 2021 **Sickle Cell Disease Challenge 1st Place Winner**, National Heart, Lung, and Blood Institute
Designed a trivia game prototype on Figma to help patients learn about Sickle Cell (\$25,000)
 - 2021 **Publication Recognition Award**, CU Boulder, Computer Science Department (\$200)
 - 2021 **Collegiate Award Finalist**, National Center for Women in Technology
 - 2020 **Department Summer Research Fellowship for Outstanding PhD TA**, CU Boulder,
Computer Science Department (\$6,000)
 - 2020 **Chateaubriand Fellow**, Embassy of France in the United States (€7,298)
 - 2020 **Graduate Research Fellow (NSF GRFP)**, National Science Foundation (\$114,000)
 - 2020 **Collegiate Award Finalist**, National Center for Women in Technology
 - 2020 **Award Winner for Aspirations in Computing**, National Center for Women in Technology
 - 2019 **Graduate Student Research Competition Winner**, ACM ASSETS 2019 (\$500)
1st Place Winner at the ACM Graduate Student Research Competition [P.2]
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Publications

Peer-Reviewed Conference Papers

- [C.6] Wenhan Xue, Kelly Ramos Pereira, Taniyah Hicks, Maitraye Das, **Vinitha Gadiraju**. 2026.
Navigating New Disability as a Sociotechnical Process. In Proceedings of the 2026
SIGACCESS Conference on Computers and Accessibility (ASSETS '26). ACM. To be published.

- [C.5] **Vinitha Gadiraju**, Lucia Jayne, Shaun K. Kane. 2024. "It's an independent living skill, but covered with fun!": Prompting At-Home Skill Development for Children with Vision Impairment. In *Proceedings of the 2024 SIGACCESS Conference on Computers and Accessibility (ASSETS '24, 28% acceptance rate)*. ACM. DOI: <https://doi.org/10.1145/3663548.3675626>
- [C.4] **Vinitha Gadiraju**, Shaun Kane, Sunipa Dev, Alex Taylor, Ding Wang, Robin Brewer, and Emily Denton. 2022. "I wouldn't say offensive but...": Disability-Centered Perspectives on Large Language Models. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*. ACM. DOI: <https://doi.org/10.1145/3593013.3593989>
- [C.3] **Vinitha Gadiraju**, Jérémie Garcia, Shaun K. Kane, Anke M. Brock. 2021. "It is fascinating to make these beasts fly": Understanding Visually Impaired People's Motivations and Needs for Drone Piloting. In *Proceedings of the 2021 SIGACCESS Conference on Computers and Accessibility (ASSETS '21, 29% acceptance rate)*. ACM. DOI: <https://doi.org/10.1145/3441852.3471219>
- [C.2] **Vinitha Gadiraju**, Olwyn Doyle, and Shaun K. Kane. 2021. Exploring Technology Design for Students with Vision Impairment in the Classroom and Remotely. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21, 26.3% acceptance rate)*. ACM. DOI: <https://doi.org/10.1145/3411764.3445755>
- [C.1] **Vinitha Gadiraju**, Annika Muehlbradt, and Shaun K. Kane. 2020. BrailleBlocks: Computational Braille Toys for Collaborative Learning. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20, 24.3% acceptance rate)*. ACM. DOI: <https://doi.org/10.1145/3313831.3376295>

Journal Articles

- [J.1] Amy Ouyang, Manasa Gadiraju, Veda Gadiraju, Landon Power, **Vinitha Gadiraju**, Gloria Liu, Kristin P. Williams, Michael M. Binkley, Sherif M. Badawy, and Melanie E. Fields. "GRAPES: Trivia game increases sickle cell disease knowledge in patients and providers and mitigates healthcare biases." *Pediatric Blood & Cancer* 69, no. 7 (2022): e29717.

Workshop Papers

- [W.1] Tiffany Horter, Serena Booth, **Vinitha Gadiraju**. 2024. Helping People Predict Agent Behaviors by Operationalizing the Variation Theory of Learning. *NeurIPS 2024 Workshop on Behavioral Machine Learning*. <https://openreview.net/forum?id=1mO4BuJowX>

Posters, Extended Abstracts, and Presentations

- [P.5] Wenhan Xue, Manasa Kudumu, Kayla Mullen, Sneha Sriram, Alicia Boyd, **Vinitha Gadiraju**. 2025. "Characterizing Uses and Prompting Strategies of LLM-Based Chatbots Among

Neurodivergent Individuals. 2025 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '25). ACM.

- [P.4] Kayla Mullen, Wenhan Xue, Manasa Kudumu. Advised by **Vinitha Gadiraju**. 2024. "I'm treating it kind of like a diary": Characterizing How Users with Disabilities Use AI Chatbots. Student Research Competition (2nd Place Winner) at the 2024 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '24). ACM. DOI: <https://doi.org/10.1145/3663548.3688549>
- [P.3] **Vinitha Gadiraju**. 2021. Tactile Play and Literacy Learning for Visually Impaired Children. Doctoral Consortium at the 2021 *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '21). ACM.
- [P.2] **Vinitha Gadiraju**. 2019. BrailleBlocks: Braille Toys for Cross-Ability Collaboration. In *Proceedings of the 2019 International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS '19), (pp. 688-690). ACM. DOI: <https://doi.org/10.1145/3308561.3356104>
- [P.1] **Vinitha Gadiraju** and Shaun K. Kane. 2019. Computational Tools to Enable Learning Braille. Poster at CRA-W Grad Cohort for Women.
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Invited Talks

- 2026 Union College (Computer Science Seminar Speaker Series)
From Representation to Reality: How Technology, AI, and Disability Shape Each Other
- 2025 University of Toronto (COMPASS Accessibility Panel)
- 2025 University of Maryland (BBL Speaker Series)
Title: Navigating Bias and Leveraging AI: Exploring the Dual Reality for Users with Disabilities
- 2024 Northeastern University (Guest Lecture)
Title: Participatory Design Methods to Investigate Accessible Tools and Disability Representation
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Academic Research

Principal Investigator, Gadiraju Lab

Wellesley College, July 2023 - Current

- Investigating and designing perinatal information tools to contextually support shared decision making between pregnant patients and their clinicians.
(Submitted for 2026 NSF CAREER Award)
 - Built community partnerships and conducted preliminary grounding interviews with *Partner to Decide*, a nonprofit supporting perinatal decision-making, and *Bump and Beyond*, a pregnancy education initiative at Boston Medical Center

- Interviews contextualized patient information-seeking within perinatal care for socioeconomically, linguistically, and culturally diverse populations. These collaborative sessions also resulted in design requirements for context-adaptive information systems
 - Conducted two workshops with the Patient Advisory Board members from *Bump and Beyond* and *Partner to Decide* to better characterize the decision types, timelines, and facets of decision-making that patients require information and support with during pregnancy.
 - Currently developing methodology for patient workshop sessions to understand how contextual information presentation adaptations in technology (interaction modality, depth, organization, patient preparation, and accessibility) could better support decision making for pregnant people.
- Capturing how couples use AI to mediate their long-term cohabiting romantic relationships and characterizing tangible relationship impacts as a result. (In collaboration with Dr. Calvin Liang at Northwestern University, Dr. Dhruv Jain at University of Michigan and Dr. Taslima Akter at University of Texas San Antonio).
 - Interviewed 14 participants in relationships to understand mediation roles of AI chatbots, why people turn to AI to take on these roles, how people conceptualize AI within these roles, and resulting impacts on the relationship.
 - Analysis will inform a longitudinal study documenting shared couple perspectives and real-time reflection as AI mediated situations arise within relationships.
- Translating mental model visualizations to a digital tool to support adolescents in Dialectical Behavior Therapy (DBT). (In collaboration with Dr. Yaniv Yacoby at Wellesley College).
 - Surveying adolescents (college students) for individual mental models to capture the diverse ways in which people process their emotional state changes.
 - Running focus groups with small participant groups to understand how they discuss and converge on common visualizations to represent emotional and mental state changes using a design tool (FigJam).
 - Working with Symbolic Modeling experts James Lawley and Penny Tompkins to develop a “clean language” methodology to elicit unbiased visualizations
 - Developing a prototype application for adolescents in therapy to build mental model visualization using the identified metaphor elements.
- Investigated how recently disabled adults use technology to mediate diagnosis, health management, and lifestyle adaptations. (In collaboration with Dr. Maitraye Das at Northeastern University).
 - Conducted semi-structured interviews with 14 recently diagnosed disabled adults to capture challenges navigating the medical system, support structures, technology use within this process, and identity formation [C.6].
- Characterized Large Language Model use among neurodivergent people within their daily routines [P.4].

- Interviewed neurodivergent adults to characterize how they used generative AI tools in their daily routines and within their broader technological ecosystem.
- Conducted diary studies to build a contextual conversational dataset between people with disabilities and chatbots, while highlighting challenges, benefits, and harms during interaction.

Graduate Researcher, Superhuman Computing Lab

CU Boulder, Fall 2018 - May 2023

- Investigated collaborative Independent Living Skill Development for blind or visually impaired (BVI) children and their parents at home.
 - Conducted an interview and diary study to characterize how families develop independence at home, what additional skills they value, and how technology can support development in these multifaceted areas.
- Led an ethnographic study at the Colorado School for the Deaf and the Blind to understand educator values for BVI children, current technology use for instruction, and how game-based technology can support instructors in the classroom and remotely [C.2].
 - Observed learning, teaching, and collaboration strategies between peers and educators related to Braille education, literacy, and Expanded Core Curriculum.
 - Conducted informal interviews with students, educators, and school administration.
 - Co-designed with educators to create games and activities to teach literacy and social skills for remote education and in-person instruction.
- Designed and created BrailleBlocks, a learning tool using tangible blocks and games to help visually impaired children and sighted collaborators learn, teach, and practice Braille [C.1].
 - Built tangible blocks representing Braille cells and coded an interface with corresponding educational games.
 - Programmed with OpenCV to color detect pegs in the blocks for real-time Braille letter and word detection.
 - Conducted user studies with sighted parents and visually impaired children to evaluate the system and elicit feedback.

Chateaubriand Research Fellow

École nationale de l'aviation civile, Spring 2021 - Summer 2021

- Led two projects to explore accessible drone piloting for BVI adults.
 - Surveyed and interviewed visually impaired participants to understand their motivation to fly and suggestions for how piloting can be made accessible [C.3].
 - Built a wearable accessible drone piloting system with audio and haptic feedback to indicate direction and depth during flight.
 - Conducted participatory design based user studies with visually impaired adults for early evaluation of the prototype.
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Industry Experience

Research Intern, Google Research - People + AI Research

Mountain View, California, Summer 2022

- Led a project on characterizing Large Language Model harm and bias towards people with disabilities by facilitating interaction between participants and an associated chatbot [C.4]
 - Designed AI prompts to moderate conversations about disability between participants and the language model.
 - Moderated focus groups with 56 participants identifying across diverse disabilities.
 - Conducted iterative thematic analysis to generate a codebook for open coding.

Teaching

Spring 2026	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2025	Introduction to Programming (CS111), Wellesley College Designing for Accessibility (CS325), Wellesley College
Spring 2025	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2024	Introduction to Programming (CS111), Wellesley College Designing for Accessibility (CS325), Wellesley College
Spring 2024	Introduction to Programming (CS111), Wellesley College
Fall 2023	Introduction to Programming (CS111), Wellesley College Human-Computer Interaction (CS220), Wellesley College
Fall 2021	Course Manager Fundamentals of Human-Computer Interaction (CSCI 3200), CU Boulder
Fall 2018 - Spring 2020	Teaching Assistant Fundamentals of Human-Computer Interaction (CSCI 3200), CU Boulder
Spring 2017	Program Manager and Tutor Top Research Methods in Psychology (PSY 303), UOregon
Fall 2017	Tutor Top Research Methods in Psychology (PSY 303), UOregon

Mentorship

While at Wellesley College

Independent Study/Lab Members

Spring 2026 Wenhan Xue, Manasa Kudumu, Emmie Regan, Taniyah Hicks, Ivy Wang, Achieng Mellisa, Imaan Jivraj, Yuchen Xiao, Shale Tenerelli
Fall 2025 Sneha Sriram, Wenhan Xue, Manasa Kudumu, Emmie Regan, Rachel Claire Henry, Taniyah Hicks, Ivy Wang, Achieng Mellisa
Spring 2025 Sneha Sriram, Wenhan Xue, Manasa Kudumu, Kamora Adejo, Emmie Regan
Fall 2024 Sneha Sriram, Wenhan Xue, Manasa Kudumu
Spring 2024 Sneha Sriram, Wenhan Xue

Honors Thesis Advisees

2023-2024 Tiffany Horter
CS Thesis: Getting In Your Robot's Head: Building Mental Models of Robots Using Human-Concept Learning [W.1]

Honors Thesis Committees

2025-2026 Karen Xiao
CS Thesis:

Aubrey Cantrell
CS Thesis:

2024-2025 Janavi Padala
CS Thesis: It's Time To Chime In! A Video Game on Political Misinformation and Individual Impact

Anna Zhou
CS Thesis: Investigating How Technologies and Platforms Shape the VTuber Experience

Maya Mau
CS Thesis: Building and Assessing a Web-based, Accessible, Three-Dimensional Modeling Tool to allow High School Students to Explore Molecular Structure, Intermolecular Interactions, and Molecular Dynamics Simulations

2023-2024 Zhihan Xu
CS Thesis: The Mysteries of Large Language Models: Tracing the Evolution of Transparency in OpenAI's GPT Models

Summer Research Advising

2026 Alison Ingalls

Kalei Kieu
Project: AI Mediation in Romantic Relationships

Yuchen Xiao
Ziyu Li
Project: Perinatal Decision Making Support

Anisa Rodriguez
Achieng Mellisa
Project: Mental Model Visualizations for DBT

2025 Taniyah Hicks
 Kelly Ramos Pereira
 Wenhan Xue
Project: Navigating Recent Disability Diagnosis

Ivy Wang
Achieng Mellisa
Project: Mental Model Visualizations for DBT

2024 Wenhan Xue
 Manasa Kudumu
 Kayla Mullen
Project: Disability Representation in Large Language Models

Other Student Mentoring

2024 Jami Johnson, Web Development Advising

Non-Wellesley Research Advising

2024-2025 Akankshita Dash, Graduate Student at Stanford University
Project: Wikipedia Image Descriptions for BLV Users

2024 Sarah Adams, Undergraduate Student at Harvard University
Project: Clinician Perspectives on ML in suicide risk prediction

2024 Sydney Levy, Graduate Student at Harvard University
Project: Clinician Perspectives on ML in suicide risk prediction

2023-2025 Alicia Boyd, Postdoctoral Researcher at New York University
Project: Disability Representation in Large Language Models

While at CU Boulder

2022-2023 Lucia Jayne - M.S. in Computer Science (CU)

2022 Alexis Grémont - M.S. in Aeronautics (École nationale de l'aviation civile)

Service

2026	Program Committee, ACM FAccT '26, Montréal, Canada
2026	Hybrid Chair, ACM ASSETS '26, Porto, Portugal
2025	Reviewer, ACM CHI '26, Barcelona, Spain
2025	Hybrid Chair, ACM ASSETS '25, Denver, USA
2025	Program Committee, ACM ASSETS '25, Denver, USA
2024	Area Chair, Accessibility and Aging Track, ACM CHI '25, Yokohama, Japan
2024	Juror, Experience Reports, ACM ASSETS '24, St. John's, Newfoundland and Labrador
2024	Reviewer, ACM CSCW '24, San José, Costa Rica
2024	Reviewer, ACM IDC '24, Delft, Netherlands
2024	Program Committee, ACM FAccT '24, Rio De Janeiro, Brazil
2023	Area Chair, Accessibility and Aging Track, ACM CHI '24, Honolulu, USA
2023	Reviewer, ACM ASSETS '23, New York, USA
2022	Late-Breaking Work Program Committee, ACM CHI '23, Hamburg, Germany
2022	Reviewer, ACM CHI '23, Hamburg, Germany
2022	Internship Guide Panelist, CU Boulder
2021	Teaching Assistant Hiring Interviewer, CU Boulder
2021	Late-Breaking Work Program Committee, ACM CHI '22, New Orleans, USA/Virtual
2021	Student Volunteer, ACM ASSETS, Virtual
2020	Prospective Faculty Student Host, CU Boulder
2020	Graduation Application Feedback Program Mentor, CU Boulder
2020	Student Volunteer, ACM CHI (Canceled due to COVID-19), Honolulu, USA
2019	Reviewer, ACM/IEEE HRI '20, Cambridge, UK

Organizations and Departmental Service

Environmental Sustainability Committee

Wellesley College, Fall 2024 – Present

- Member of Advisory Board to Wellesley College on fostering a more sustainable and waste minimizing campus. Sub-committee focused on reducing dining hall waste.

WOC Faculty Mentoring Group

Wellesley College, Fall 2023 – Present

- Professional support/development group for tenure-track women of color at Wellesley.

Graduate Committee and Graduate Student Association

CU Boulder, Department of Computer Science, Fall 2021 – Spring 2022

- Communicated and addressed departmental issues between faculty (Graduate Committee) and graduate student leaders (Graduate Student Association)
- Advocated for fair tuition for internship credits for international students.
- Planned inclusive, unifying departmental events, such as trivia nights and scavenger hunts.

Graduate Committee and Graduate Student Association

CU Boulder, Department of Computer Science, Fall 2021 – Spring 2022

- Communicated and addressed departmental issues between faculty (Graduate Committee) and graduate student leaders (Graduate Student Association)
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- Planned inclusive, unifying departmental events, such as trivia nights and scavenger hunts.

Antiracism and Inclusion Committee

CU Boulder, Department of Engineering, Summer 2020 – Winter 2020

- Worked with administration to analyze student survey data and faculty/course review questionnaires to identify issues with inclusivity, racism, and harassment in the department.
- Advocated for and helped implement inclusive recruitment and retention practices for students and faculty.
- Designed a survey to characterize the current racial climate and culture of the department.

Graduate Student Advisory Board, Computer Science Representative

CU Boulder, Department of Engineering, Fall 2019 – Fall 2020

- Planned department-wide social events, such as mixers and virtual game nights.